

Teorema de Gram-Schmidt

Teorema 1 (Gram-Schmidt). Sea $\{u_n\}_{n \in \mathbb{N}}$ un conjunto linealmente independiente y numerable en un espacio prehilbert H

$$\implies \exists \{v_n\}_{n \in \mathbb{N}} \text{ un sistema ortonormal : } \forall n \in \mathbb{N} : \text{span}(\{u_i\}_{i=1}^n) = \text{span}(\{v_i\}_{i=1}^n).$$

Demostración: Esta demostración será constructiva. Definimos $q_1 = u_1$ y $v_1 = \frac{q_1}{\|q_1\|}$. Definimos $q_2 = u_2 - \alpha_2 q_1$ de tal forma que $q_2 \perp q_1$, es decir, $\langle q_2, q_1 \rangle = 0$. Calculamos α_2 :

$$0 = \langle u_2 - \alpha_2 q_1, q_1 \rangle = \langle u_2, q_1 \rangle - \alpha_2 \langle q_1, q_1 \rangle \implies \alpha_2 = \frac{\langle u_2, q_1 \rangle}{\|q_1\|^2}.$$

Luego, $q_2 = u_2 - \frac{\langle u_2, q_1 \rangle}{\|q_1\|^2} q_1$ y definimos $v_2 = \frac{q_2}{\|q_2\|}$. Iterando este proceso, suponemos que hemos definido $\{q_i\}_{i=1}^{k-1}$ ortogonales. ■

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